

**LOMBA KOMPETENSI SISWA (LKS)
SEKOLAH MENENGAH KEJURUAN
TINGKAT PROVINSI DKI JAKARTA TAHUN 2024**

**TEST PROJECT
INFRASTRUCTURE PROGRAMMABILITY & AUTOMATION**



**Bidang Lomba
IT NETWORK SYSTEMS ADMINISTRATION**

Introduction

Accurate and up-to-date documentation has always been a challenge in IT. With multiple engineers working on the same systems, it is hard to keep track of who changed what. Applix Corporation decided to fix this problem and hired you to modernize, harden, and extend their infrastructure.

Description of Project and Tasks

You will be migrating VMs to Infrastructure as Code (IaC) and simplifying the process of creating new services. You have access to development VMs (DEV-LIN). These VMs can be used for developing and testing your work.

Login for all VMs and Devices:

Linux

Username : root/user

Password : P@ssw0rd

Windows

Username : administrator/user

Password : P@ssw0rd

Instructions to the Competitor

Part 0. Basic Configuration

Install and configure all VMs according to the appendix.

Part 1. Linux

Use Ansible to configure the Linux hosts LIN[1-2] from BASTION VM. Create file `/home/user/workfolder/hosts` which will be used as the ansible inventory. For marking, all playbooks will be run in order using the command `ansible-playbook -i linux/playbookname.yml` in the `/home/user/workfolder` directory as user **user**.

Variables like `hostname`, `webport` and `webcolor` in `/home/user/workfolder/hosts` are subject to change for marking.

General

- Create a directory `/home/user/workfolder/linux`
 - Place all playbooks directly inside this directory.
 - You are free to create folders/files in this directory for running the playbooks.
 - When you run the playbooks for the second time, every task should either be in the “ok” state or be “skipped.”

Hostname

- Create a playbook called **1-hostname.yml** for configuring the hostname
 - Every host should be assigned a hostname, derived from the `hostname` variable found in the `/home/user/workfolder/hosts` file.

DNS

- Create a playbook called **2-dns-server.yml** for configuring at least two DNS servers
 - Install a DNS service (**Bind9**) on hosts in the group “dns”.
 - Create zone domain `linux.com`.
 - Create A record for `linux.com` to `10.0.10.11`.
 - Ensure every host in the group Linux has an A record for `<hostname>.linux.com`.
- Create a playbook called **3-dns-client.yml**
 - Configure DNS nameservers to `10.0.10.11` and `10.0.10.20` for all group “linux”.

WEB Server

- Create a playbook called **4-web-server.yml** for configuring two or more web servers (**nginx**)
- Install a web service on all hosts in the group “web”
 - The local website should listen on port `webport` variable in `/home/user/workfolder/hosts` called `<hostname>.linux.com`.
 - Display the following content with text color based on the `webcolor` variable in `/home/user/workfolder/hosts` file

`"Hello from <hostname> !"`

Users

- Create a playbook called **5-users.yml** for importing users on all LIN hosts
 - Create user-xxxxx where xxxxx is 00001-99999. Password should be the same as the username
 - Make sure, that the password is not changed if there is already an existing user with the same username and UID

Part 2. Windows

Use Ansible for configuring the Windows hosts, labeled as WIN, from the BASTION VM. Create file `/home/user/workfolder/hosts` which will be used as the ansible inventory. For evaluation purposes, all playbooks will be executed in sequence using the command “`ansible-playbook -i hosts windows/playbookname.yml`” within the `/home/user/workfolder` directory as user **user**.

General

- Create a directory **/home/user/workfolder/windows**
 - All playbooks should be located at the root of this directory
 - You are free to create folders/files in this directory for running the playbooks
 - When you run the playbooks for the second time, every task should either be in the “ok” state or be “skipped.”

Hostname

- Create a playbook called **1-hostname.yml** for configuring the hostname
 - All hosts should receive the hostname based on the “hostname” variable in `/home/user/workfolder/hosts` file

Security and logging

- Create a playbook called **2-sec-log.yml** for configuring security settings
 - Stop and disable the RDP Service on all Windows hosts

DNS

- Create a playbook called **3-dns-server.yml** to configure DNS Server
 - Install DNS Server on WIN
 - Create Zone windows.com set as primary
 - Create A record for windows.com to 10.0.10.20.
 - Create A record for <hostname>.windows.com.
 - Create A record for www.windows.com to 10.0.10.20
- Create a playbook called **4-dns-client.yml**
 - Configure DNS nameservers to 10.0.10.20 and 10.0.10.11 for all group “windows”.

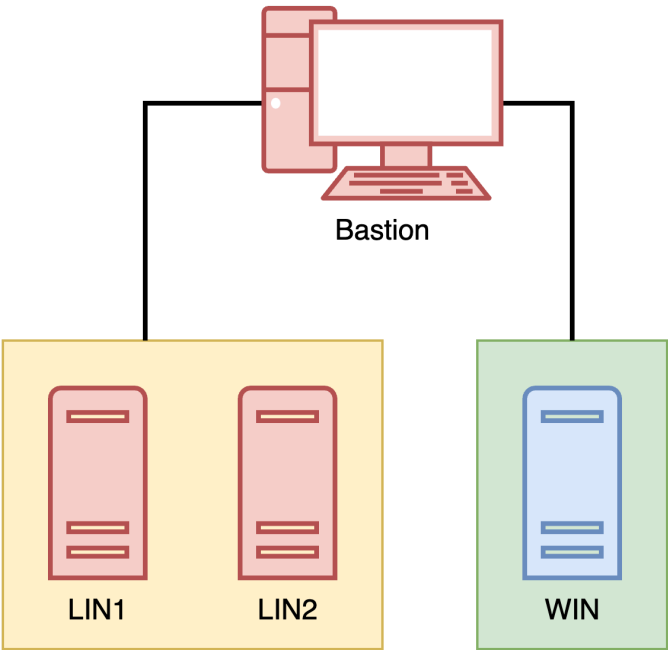
WEB Server

- Create a playbook called **5-web-server.yml** for installing the web server
 - Install the IIS WEB Server feature in the group windows
- Create http only for www.windows.com.
 - Display the following content.

“Hello from <hostname> !”

Appendix

Topology



System table

ID	IP	OS	DESCRIPTION
BASTION	10.0.10.1	Debian 12.4 (GUI)	Main machine to run ansible playbook
LIN1	10.0.10.11	Debian 12.4 (CLI)	Dynamic Config
LIN2	10.0.10.12	Debian 12.4 (CLI)	Dynamic Config
WIN	10.0.10.20	Windows Server 2022 (Desktop)	Dynamic Config